- Cover
- Contents
- Introduction
- Reflective Journals
- Career Development Plan
- Curriculum Vitae
- Evidence of a Skill Course

PORTFOLIO | CAREER PREPARATION FOR UNDERGRADUATES | BATCH: E/2021

Dilumi Gimansha Karunarathna

BSc (Hons) in Information Technology
Specializing in Information Technology - Final Year

B.Sc. Hons in Information Technology	Undergraduate Portfolio
Specialization	Career Stage
Software Engineering	Preparing for Professional Practice
Batch	Graduation Year
E/21 - Weekday Undergrad	2026 (Expected)
Current Location	Portfolio Version
Kandy, Sri Lanka	March 2026

dilumigimansha.github.io/dilum-portfolio

Table of Contents

Introduction 3
Reflective Journals 4
10.1 – Academic Writing 4
10.2 – Presentation Skills 5
10.3 – Report Writing 6
10.4 – Formal and Informal Letters 7
10.5 – Memo Writing 8
10.6 – Professional Email Writing 9
Career Development Plan 10
11.1 – Personal SWOT Analysis 10
11.2 – Career Goals 11
Curriculum Vitae 12
Evidence of a Skill Course 13

Introduction

Hi, I'm Dilumi. I'm **Academic Cayenge Direct Enterprise Karunarathna**, and I am fully delighted to introduce myself as a part of **Undergraduate Portfolio** for the **Bachelor of Science (Hons) in Information Technology**. I am currently specializing in **Software Engineering** and I am in the **Weekday Undergrad** batch. I am from **Kandy, Sri Lanka**.



My academic journey began at **Sujatha Vidyalaya, Matara**, a school that laid a strong foundation for my studies. Currently, I am pursuing my undergraduate degree at **University of Moratuwa**, where I am expanding my knowledge and skills in my chosen field.

I have chosen to follow the **Bachelor of Science (Hons) in Information Technology** because it aligns perfectly with my passion for technology and my career aspirations. This program offers a comprehensive curriculum that covers a wide range of topics, including programming, software development, data management, and network security. I am excited about the opportunity to learn from experienced professors and industry professionals, and I look forward to applying my knowledge to real-world projects.

My interest in Academic Journey stems from my desire to stay at the forefront of technological advancements. In today's rapidly changing world, it is essential to continuously learn and adapt to new technologies. I am confident that the skills, knowledge, and values I have developed during my undergraduate studies will enable me to make a significant contribution to the field of Information Technology.

I am confident that the skills, knowledge, and values I have developed during my undergraduate studies will enable me to make a significant contribution to the field of Information Technology. I am excited about the opportunity to learn from experienced professors and industry professionals, and I look forward to applying my knowledge to real-world projects.

As a student pursuing the degree of **Bachelor of Science (Hons) in Information Technology**, I am fully dedicated to my studies and am committed to achieving excellence in all my academic endeavors. I am confident that my skills, knowledge, and values I have developed during my undergraduate studies will enable me to make a significant contribution to the field of Information Technology.

Reflective Journals

Academic Writing

- Throughout my academic life, I have developed a strong foundation in academic writing, which has been instrumental in my success as a student. I have learned to write clear, concise, and well-organized essays, reports, and research papers that effectively communicate my ideas and arguments. I have also developed a strong ability to critically analyze and evaluate information from various sources, which has been essential for my academic success.
- One of the most important skills I have developed is the ability to write in a formal and objective manner. I have learned to avoid using personal pronouns and to instead focus on presenting facts and evidence in a logical and persuasive manner. This skill has been particularly useful in my scientific writing, where I am required to write in a precise and objective manner.
- I also learned about the main elements of academic writing, such as the thesis statement, discussion, conclusion, and recommendation. I learned how to write a strong thesis statement that clearly states the purpose of my paper and how to support it with evidence and arguments. I also learned how to write a conclusion that summarizes my findings and suggests areas for further research.
- Overall, I have found academic writing to be a challenging but rewarding experience. It has helped me to develop my critical thinking skills, improve my communication skills, and prepare me for my future career. I am confident that the skills I have learned will be valuable assets in my professional life.

Presentation Skills

- Before this session, presentations often made me feel anxious and unprepared. I struggled with organizing my thoughts and delivering them confidently. The materials on AI tools, especially Gamma, have been incredibly helpful in structuring my presentations effectively. I've learned how to create engaging slides that support my message without overwhelming the audience.
- I also learned about the importance of practice and feedback. Rehearsing my presentation multiple times has helped me refine my delivery and timing. Additionally, receiving constructive feedback from my peers has been invaluable in identifying areas for improvement.
- Overall, I feel much better prepared for my next presentation. I'm excited to apply what I've learned and continue developing my skills. I believe that with practice and the right tools, I can deliver presentations that are both informative and engaging.

Report Writing

- This session introduced me to the formal structure of report writing and helped me understand how reports differ from essays. Previously, I tended to write in a more narrative style, but I now realize that reports require a clear, objective, and well-organized approach. I found the breakdown of the different sections, such as the abstract, introduction, methodology, results, discussion, and conclusion, very helpful.
- I also learned about the main elements of report writing, such as the table of contents, bibliography, discussion, conclusion, and recommendations. I found the guidelines for writing each section very useful, and I plan to refer to them in my future assignments. I also learned about the importance of using a consistent referencing style, such as Harvard or APA, and I will be sure to do so.
- Overall, I found this session very informative and helpful. I feel more confident about my ability to write a report now, and I am looking forward to applying what I have learned in my future assignments. I also plan to review the materials from this session to reinforce my knowledge.

Formal and Informal Letters

- This session was very helpful in understanding the differences between formal and informal letters. I learned that formal letters are used in professional or official situations, while informal letters are used for personal communication with friends and family. I also learned about the different elements of a formal letter, such as the sender's address, the recipient's address, the date, the salutation, the body of the letter, and the closing.
- I found the examples of formal and informal letters very helpful in understanding the appropriate tone and language to use in each type of letter. For example, I learned that formal letters should be written in a polite and respectful tone, and they should avoid using contractions and slang. In contrast, informal letters can be written in a more relaxed and conversational tone, and they can include contractions and slang.
- I also learned about the importance of proofreading my letters carefully before sending them. I will make sure to check for any spelling or grammar errors, and I will also make sure that the letter is well-organized and easy to read.
- Overall, I found this session very informative and helpful. I feel more confident about my ability to write formal and informal letters now, and I am looking forward to applying what I have learned in my future correspondence.

Memo Writing

- Before the session, I had limited knowledge of memo writing. I was unaware of the specific format and the internal workplace communication purpose they serve. I found the explanation of the different sections, such as the "To," "From," "Date," and "Subject" lines, very helpful in understanding the structure of a memo. I also learned about the importance of using a clear and concise subject line to grab the reader's attention and summarize the main point of the memo.
- I found the examples of memos very helpful in understanding the appropriate tone and language to use in each type of memo. For example, I learned that memos should be written in a professional and objective tone, and they should avoid using personal pronouns and slang. I also learned about the importance of using bullet points and numbered lists to make the information easy to read and understand.
- I also learned about the importance of proofreading my memos carefully before sending them. I will make sure to check for any spelling or grammar errors, and I will also make sure that the memo is well-organized and easy to read.
- Overall, I found this session very informative and helpful. I feel more confident about my ability to write memos now, and I am looking forward to applying what I have learned in my future correspondence. I also plan to review the materials from this session to reinforce my knowledge.

Professional Email Writing

- This session helped me understand the importance of clear and concise communication in professional emails. I learned that email is one of the most common forms of communication in the workplace, so it is important to write emails that are easy to understand and get straight to the point.
- I found the tips on writing effective subject lines very useful. I will make sure to write subject lines that are specific and informative, so that the recipient knows what the email is about before they open it.
- I also learned about the importance of using a professional tone and avoiding slang and emojis. I will make sure to proofread my emails carefully before sending them, to avoid any errors in grammar or spelling.
- Overall, I found this session very helpful and informative. I feel more confident about my ability to write professional emails now, and I am looking forward to applying what I have learned in my future correspondence.

Career Development Plan

Name:	Academic Cayenge Direct Enterprise Karunarathna
Current Role:	Undergraduate, BSc (Hons) in Information Technology, SLIIT
Career Goal:	Quality Assurance Engineer
Vision:	To become a highly skilled and respected QA professional who contributes to the delivery of reliable, user-friendly software products, while growing into a senior leadership role within the quality assurance profession.

Personal SWOT Analysis

STRENGTHS
- Strong academic background in software engineering
- Familiarity with software development concepts
- Good attention to detail, communication skills (VPRM)
- Self-motivation, quick learning ability, and time management
- Effective in teamwork and team working

WEAKNESSES
- Limited practical, hands-on QA testing experience
- Lack of professional certifications in software testing and quality assurance
- Limited practical or real-life problem-solving skills
- Public speaking confidence still developing

OPPORTUNITIES
- Increasing demand for QA engineers locally & globally
- Online courses, certifications, and bootcamps available
- SLIIT alumni network & industry connections
- Remote work & part-time job opportunities

THREATS
- Competitive job market for entry-level IT roles
- Rapidly changing technologies requiring continuous learning
- AI-driven automated testing tools may reduce demand for manual testers

Short Term Goals (1 – 2 Years)

Goal 1 – Graduate with Honours
- Successfully complete the final year of SLIIT, achieving an assessment in a high standard
- Complete a research project or industry-based project in QA or testing
- Strengthen the professional profile and demonstrate a commitment to academic excellence

Goal 2 – Obtain a Recognized QA Certification
- Pursue and complete the ISTQB Foundation Level certification in Software Testing
- Validate theoretical knowledge of testing principles and test design techniques
- Gain a globally recognized qualification to boost job prospects and credibility in the software industry

Goal 3 – Secure an Entry-Level QA Engineer Position
- Actively apply for QA engineer or software tester roles in IT companies in Sri Lanka
- Target organizations that work with Agile methodologies
- Tailor CVs and cover letters to highlight testing skills and projects through VPRM
- Prepare thoroughly for technical interviews

Medium Term Goals (3 – 5 Years)

Goal 1 – Gain Hands-On Experience in Test Automation
- Move beyond manual testing and develop strong competency in automated testing frameworks
- Learn tools such as Selenium, Cypress, or Playwright (or similar)
- Work with or support a team that is transitioning to automation
- Recognize automation as an increasingly essential skill in modern QA practice

Goal 2 – Pursue Intermediate-Level Certification
- After gaining practical experience, pursue the ISTQB Advanced Level — Test Analyst certification
- Deepen technical expertise and demonstrate commitment to professional growth

Goal 3 – Build a Professional Network
- Actively engage with the QA community and attend local IT professional community
- Connect through LinkedIn, industry meetups, and SLIIT alumni networks
- Open doors to new opportunities, mentorship, and industry insights

Long Term Goals (5 – 10 Years)

Goal 1 – Progress to Senior QA Engineer or QA Lead Role
- Accumulate 5 years of solid experience in both manual and automated testing
- Aim to be recognized as a senior contributor within a QA team
- Develop leadership skills by mentoring junior testers
- Continue to refine quality standards and processes in my workplace

Goal 2A – Explore Specialization in Performance or Security Testing
- Explore performance testing using tools like JMeter
- Develop expertise in security testing to address the importance of cybersecurity in software development

Goal 2 – Contribute to the QA Community
- Develop my own QA professional training for Profit.
- Contribute to industry forums and blogs to share knowledge and experiences
- Potentially conduct workshops or training sessions for aspiring QA engineers

Curriculum Vitae

DILUMI GIMANSHA KARUNARATHNA
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Professional Summary

Undergraduate Software Engineering student with hands-on experience in full-stack development, agile methodologies, and DevOps practices. Proven track record of delivering production-ready applications including Real-time Communication Platform and Smart Campus Sensor Network. Skilled in Java, Python, JavaScript, and cloud technologies. Seeking a software engineering internship to contribute technical expertise and learn from experienced professionals in building scalable, user-centric solutions. Strong problem-solving abilities with excellent collaboration and communication skills.

Education

BSc (Hons) in Information Technology, Specialization in Software Engineering — Year 2026 – Present
SLIIT, Malabe Campus, Sri Lanka
Expected Graduation: 2026 | Current GPA: 3.72 / 4.00
Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Software Engineering, Web Technologies, Operating Systems, Computer Networks

Advanced Level — Year 2019 – 2021
Royal College, Colombo, Sri Lanka
Physical Science Stream | Z-Score: 1.8542

Work Experience

Intern, Programmer Trainee, Electrical & ICT Coordinator — December 2024 – June 2025
Sri Lanka Navy, Colombo, Sri Lanka

- Developed a web-based inventory management system using Laravel and MySQL, reducing manual tracking errors by 40% and saving approximately 15 hours per week
- Automated data entry processes using Python scripts, improving efficiency and accuracy of monthly reports
- Collaborated with a team of 5 developers using Git for version control and Agile methodologies for project management
- Presented project outcomes to senior officers and received positive feedback for implementation

Intern Engineer — Year 2023 – Present
Ceylon Electricity Board, Kandy, Sri Lanka

- Developed a real-time monitoring dashboard for substation equipment using React.js and Node.js, enabling remote access to critical operational data
- Implemented RESTful APIs to integrate sensor data from 12 substations into the central database
- Created automated daily and monthly report generation scripts in Python, reducing manual reporting time by 3 hours per week
- Assisted in troubleshooting and maintenance of SCADA systems and communication networks

Projects

Real-Time Collaboration Platform — Year 2025 – Present
Full-Stack Developer | Team of 4

- Developed a web-based platform for real-time document collaboration using WebSockets and operational transformation algorithms
- Implemented user authentication, role-based access control, and document version history
- Built responsive frontend using React.js and Tailwind CSS with dark mode support
- Deployed application on AWS EC2 with CI/CD pipeline using GitHub Actions

Smart Campus Sensor Network — Year 2024 – Present
Team Lead | Team of 3

- Designed and implemented a wireless sensor network using ESP32 microcontrollers and MQTT protocol for environmental monitoring across campus buildings
- Developed a Node-RED dashboard for visualizing temperature, humidity, and occupancy data in real-time
- Integrated machine learning model for predicting energy consumption patterns, achieving 85% accuracy
- Published project documentation and open-sourced codebase on GitHub with comprehensive setup guides

Task Management API — Year 2023 – Present
Backend Developer | Solo Project

- Built a RESTful API using Node.js, Express, and MongoDB with JWT authentication and role-based authorization
- Implemented CRUD operations, data validation, and error handling following industry best practices
- Wrote unit and integration tests using Jest and Supertest, achieving 90% code coverage
- Containerized application with Docker and created deployment documentation for easy setup

Skills & Interests

Programming Languages: Java, Python, JavaScript, TypeScript, C, SQL
Web Technologies: React.js, Node.js, Express, HTML5, CSS3, Tailwind CSS, REST APIs, WebSockets
Databases: MySQL, PostgreSQL, MongoDB, Redis
Tools & Platforms: Git, Docker, AWS (EC2, S3, Lambda), Linux, VS Code, IntelliJ IDEA, Postman
Methodologies: Agile/Scrum, Test-Driven Development, Continuous Integration/Delivery
Interests: Open-source contribution, competitive programming, technical writing, hackathons

Certifications

AWS Cloud Practitioner Essentials — September 2024
Amazon Web Services (AWS)
AWS Academy Graduate - Machine Learning Foundations — September 2024
Amazon Web Services (AWS)
Introduction to Cybersecurity — September 2023
Cisco Networking Academy

Extracurricular Activities

Software Society — Member, Organizing Committee (2022 – 2023)
Hackathon Participation — Top 10 finalist, National Level Competition (2024)
Volunteer Tutor — Mathematics and Science for Grade 9 students (2021 – 2022)

References

Dr. Ruwan Perera — Senior Lecturer, Department of Computer Science, University of Peradeniya — ruwan.perera@sci.pdn.ac.lk
Mrs. Thilini Wickramasinghe — Software Engineering Manager, Virtusa (Pvt) Ltd — thilini.w@virtusa.com

Evidence of a Skill Course

Module Code: ENG1013 | **Reference in Portfolio:** Reference in English

Grade Received	Credits Earned	Mode of Study
Good (Category_Passed)	February 2026	Communication Skill Skill

Summary

This Diploma improved my English language proficiency in writing, grammar, reading, and speaking. It strengthened my ability to communicate effectively in academic and professional contexts.

Career Relevance

It is a Quality Assurance Engineer role, communication is essential for writing clear reports, test cases, and professional emails. This qualification helps me demonstrate strong communication skills necessary in a professional environment.

Diploma in English
Issued by the Board of Study Courses
This is to certify that the student mentioned below has successfully completed the Diploma in English conducted by the Board of Study Courses
A. L. D. Gimansha Karunarathna
Index No: 2021/E/0001
Grade: Good (Category_Passed)
Date: February 2026

Module Code: ICT1013 | **Reference in Portfolio:** Higher Diploma in Information Technology

Grade Received	Credits Earned	Mode of Study
ICT Module	December 2025	Higher Diploma in Information Technology

Summary

This was a group project assessment to develop a software solution, demonstrating collaboration, teamwork, and problem-solving skills.

Career Relevance

It is an essential skill for technical roles, helping me understand software systems and collaborate effectively with developers and project teams during my internship.

SRI LANKA
INSTITUTE OF INFORMATION TECHNOLOGY
We certify that
Academic Cayenge Direct Enterprise Karunarathna
has successfully completed the course unit
DECLARE EMPLOYING
IN
INFORMATION TECHNOLOGY

Module Code: ENG1013 | **Reference in Portfolio:** AWS Cloud Web Application Builder Training

Grade Received	Credits Earned	Mode of Study
Completed (Self-Directed, 100%)	September 2025	Cloud Computing

Summary

This course provided foundational cloud concepts and the process of building and deploying web applications using AWS services.

Career Relevance

It improved my understanding of cloud infrastructure, enabling me to work more effectively in cloud-based environments.

AWS Cloud Practitioner Essentials
Issued by Amazon Web Services Training and Certification
Dilumi Gimansha Karunarathna
Completed on 15 September 2025
AWS Academy Graduate - Cloud Web Application Builder
Issued by Amazon Web Services Training and Certification
Dilumi Gimansha Karunarathna
Completed on 15 September 2025
AWS Simulacrum: Highly Available Web Applications
Issued by Amazon Web Services Training and Certification
Dilumi Gimansha Karunarathna
Completed on 15 September 2025
AWS Simulacrum: Databases in Practice
Issued by Amazon Web Services Training and Certification
Dilumi Gimansha Karunarathna
Completed on 15 September 2025

Module Code: ENG1013 | **Reference in Portfolio:** AWS Simulacrum: Highly Available Web Applications

Grade Received	Credits Earned	Mode of Study
Completed (Self-Directed, 100%)	September 2025	Cloud Computing

Summary

This simulation-based learning tool focused on building and maintaining highly available, scalable, and fault-tolerant cloud infrastructure.

Career Relevance

It enhanced my ability to design and maintain reliable IT systems, a key skill for technical roles in cloud computing.

AWS Simulacrum: Highly Available Web Applications
Issued by Amazon Web Services Training and Certification
Dilumi Gimansha Karunarathna
Completed on 15 September 2025
AWS Simulacrum: Databases in Practice
Issued by Amazon Web Services Training and Certification
Dilumi Gimansha Karunarathna
Completed on 15 September 2025

Module Code: ENG1013 | **Reference in Portfolio:** AWS Simulacrum: Databases in Practice

Grade Received	Credits Earned	Mode of Study
Completed (Self-Directed, 100%)	September 2025	Database & Cloud

Summary

This course covered database management, configuration, and performance optimization within AWS environments.

Career Relevance

It strengthened my skills in database tuning, security, and cloud-based data solutions, which are valuable for technical roles.

AWS Simulacrum: Databases in Practice
Issued by Amazon Web Services Training and Certification
Dilumi Gimansha Karunarathna
Completed on 15 September 2025